

Product overview

refrigeration potential relay service kit has become an industry standard since its introduction over fifty years ago. Its long-term reliability and easy replacement have been the key to its success. The wide calibration ranges and the universal bracket make potential relay is a must on every service truck.

Features&benefits

- ◇50A switching capability
- ◇Has a variety of mounting positions
- ◇With a variety of wiring
- ◇Dust cover type packaging structure
- ◇Used on heat pumps,vendingmachines,fans,and heating and heating and air conditioning

Product certification

c UL us	UL	UL 60947-4-1	RoHS	RoHS	RoHS Compliant

Code explanation

	PR-90	60/	Z	AA	1	(XXX)
Type						
Coil number 63/50/65/69/58/66/71/61/64						
Terminal configuration Z Metal Tab Mount D Plastic Tab Mount P Panel Mount						
Operation characteristics AA to AS (See table for operating characteristics)						
Operation position 1/2/3/4/5/6						
Special code1 XXX: Customer special requirement						

Product model and specification

OPERATING CHARACTERISTICS AT 60Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

Coil number		63		50		65		69		58		66		71		61		64	
Vmax at 40°C (VAC)		170		332		336		375		379		395		420		450		500	
Resistance at 25°C (Ω) ±10%		1500		5600		5700		7500		7600		10000		10700		11000		13800	
	H.P.U.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.
AA	120-130	111-124	20-45																
AB	130-140	120-134	20-45	132-148	40-90														
AC	150-160	130-144	20-45	140-153	40-90														
AD	160-170	140-153	20-45	150-163	40-90														
AE	170-180	149-163	20-45	162-175	40-90														
AF	180-190			171-184	40-90	162-175	40-90					180-195	40-105						
AG	190-200			180-193	40-90	171-184	40-90	180-195	40-105			189-205	40-105	180-195	60-121				
AH	200-220			186-215	40-90	180-193	40-90	190-215	40-105	180-195	40-105	186-214	60-130	189-224	60-121				
AI	220-240			205-234	40-90	186-215	40-90	208-239	50-110	190-215	40-105	204-233	60-130	204-233	60-121				
AL	240-260			224-252	40-105	205-234	40-90	224-252	50-110	208-239	50-110	223-252	60-130	223-259	60-121	223-252	60-130		
AM	260-280			243-271	40-105	224-252	40-105	239-270	50-110	224-252	50-110	242-272	60-140	242-272	60-121	239-268	60-135	239-268	60-135
AN	280-300					243-271	40-105	260-289	50-110	239-270	50-110	262-290	60-140	262-290	60-121	258-287	60-135	258-287	60-135
AO	300-320									260-289	50-110	280-310	60-140	280-310	60-121	277-305	60-135	277-305	60-135
AP	320-340											300-328	60-140	300-328	60-121	295-324	60-135	295-324	60-135
AQ	340-360													318-347	60-121	314-342	60-135	314-342	60-135
AR	350-370																	323-352	60-135
AS	360-380																		

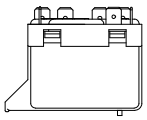
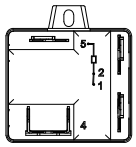
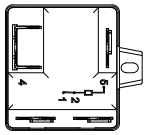
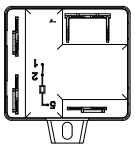
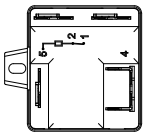
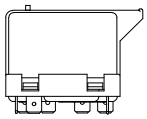
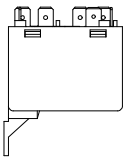
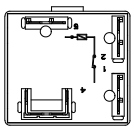
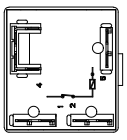
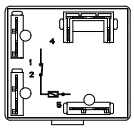
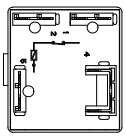
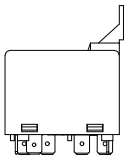
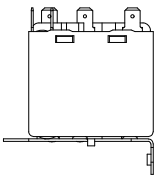
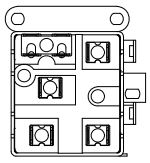
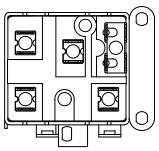
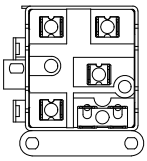
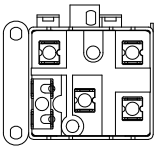
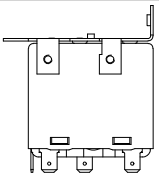
OPERATING CHARACTERISTICS AT 50Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

Coil number		63		50		65		69		58		66		71		61		64	
Vmax at 40°C (VAC)		151		299		300		338		340		356		379		400		450	
Resistance at 25°C (Ω) ±10%		1500		5600		5700		7500		7600		10000		10700		11000		13800	
	H.P.U.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.
A	120-130	111-124	20-45																
B	130-140	120-134	20-45	132-148	40-90														
C	150-160	130-144	20-45	140-153	40-90														
D	160-170	140-153	20-45	150-163	40-90														
E	170-180	149-163	20-45	162-175	40-90														
F	180-190			171-184	40-90	162-175	40-90					180-195	40-105						
G	190-200			180-193	40-90	171-184	40-90	180-195	40-105			189-205	40-105	180-195	60-121				
H	200-220			186-215	40-90	180-193	40-90	190-215	40-105	180-195	40-105	186-214	60-130	189-224	60-121				
I	220-240			205-234	40-90	186-215	40-90	208-239	50-110	190-215	40-105	204-233	60-130	204-233	60-121				
L	240-260			224-252	40-105	205-234	40-90	224-252	50-110	208-239	50-110	223-252	60-130	223-259	60-121	223-252	60-130		
M	260-280			243-271	40-105	224-252	40-105	239-270	50-110	224-252	50-110	242-272	60-140	242-272	60-121	239-268	60-135	239-268	60-135
N	280-300					243-271	40-105	260-289	50-110	239-270	50-110	262-290	60-140	262-290	60-121	258-287	60-135	258-287	60-135
O	300-320									260-289	50-110	280-310	60-140	280-310	60-121	277-305	60-135	277-305	60-135
P	320-340											300-328	60-140	300-328	60-121	295-324	60-135	295-324	60-135
Q	340-360													318-347	60-121	314-342	60-135	314-342	60-135
R	350-370																	323-352	60-135
S	360-380																		

If a 60Hz model is used at 50Hz, its 50Hz PU will be about 13% lower.
If a 50Hz model is used at 60Hz, its 60Hz PU will be about 13% higher.

Operating position

	1	2	3	4	5	6
PLASTIC TAB MOUNT						
PANEL MOUNT						
METAL TAB MOUNT						

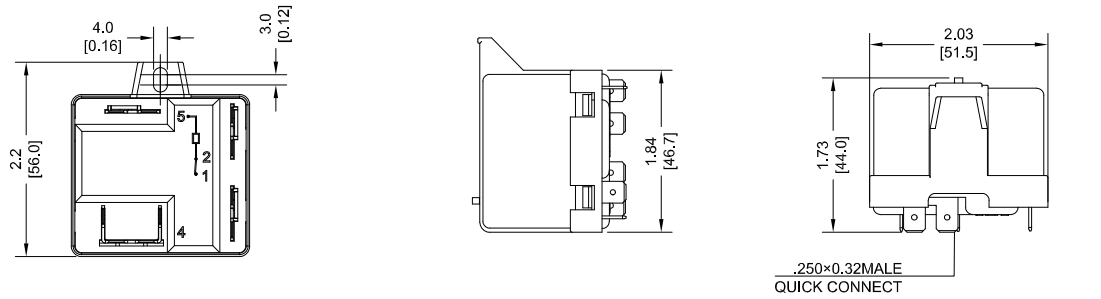
Product Characteristics

Maximum operating voltage		500VAC @-40℃~50℃
		500VAC @-40℃~85℃
Contact form		SPST-NC
Contact resistance		≤50mΩ(@24V/1A)
Rated load		35A 400VAC
Maximum commutation voltage		500VAC
Maximum commutation current		50A
Input power	Holding power	≤5 VA
	Input power	≤21.5 VA
Mechanical durability		$1X10^6$
Electrical durability		SPST-NC: 5 x 10^5OPS 16A on and off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off SPST-NC: 2 x 10^5OPS 35A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off SPST-NC: 1 x 10^5OPS 50A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off
Insulation resistance		500MΩ（500VDC）
Shock（stability）		$98m/s^2$
Vibration		10Hz to 55Hz, 1.5mm double amplitude
Humidity range		5% to 85%RH
Temperature range		-40℃ to 85℃
Lead out form		Quick connect
Weight		About 98g
Packaging method		Dust cover type

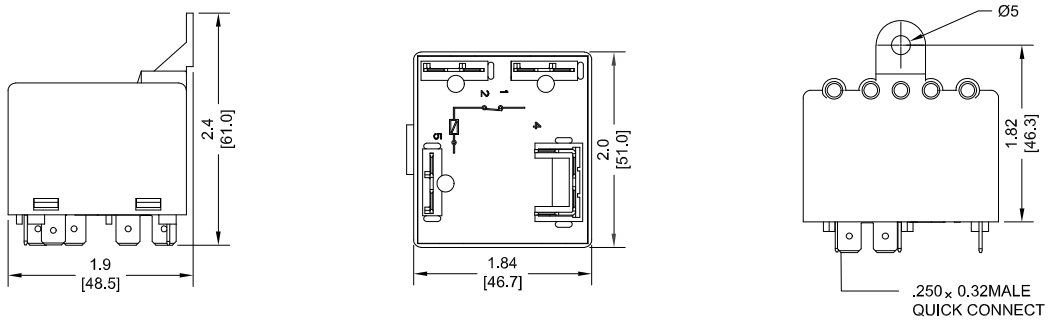
Dimensions

Plastic Tab Mount

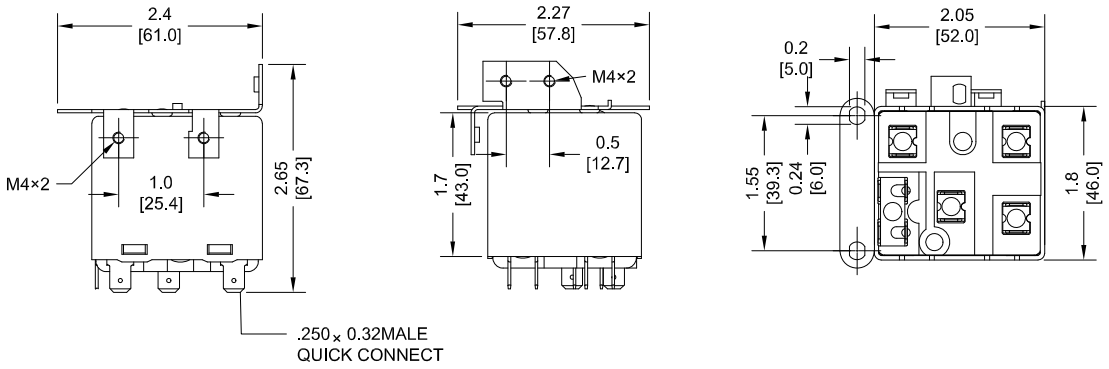
inch[mm]



Panel Mount



Metal Tab Mount



Wiring diagram

